Calibrated, domain-guarded switching for layered excited-state potentials: stable tungsten cascades and the extrapolative origin of their reduced damage

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Abstract

Layered excited-state potentials model electronic excitation in radiation-damage cascades by blending, atom by atom, the forces of a ground-state and an electronically hot machine-learned potential. In tungsten they have been reported to reduce surviving damage, but they become unstable under aggressive switching. Everything physical passes through the switching law that maps the local ionic temperature to a blend fraction. We derive that law from stated requirements. A Sommerfeld-limited map of electronic temperature is calibrated against held-out DFT at $T_e = 0.1\,\mathrm{eV}$. A quasi-static two-temperature closure gives the electronic temperature, with a noise floor measured from the estimator. A neighbourhood trust guard keeps atoms off the excited layer wherever that layer is outside its training domain. The calibration lowers the held-out force error by 13% and both integrated errors. It cannot lower energy and force errors together at $0.1\,\mathrm{eV}$, because the two fits' errors exceed the physical signal. An exact energy ledger shows that force blending heats cascades by 6–8% of the recoil energy. It is therefore not a form of electronic stopping. The instability is a collapse into a spurious dense state of the excited potential, $29\,\mathrm{eV/atom}$ below its own crystal. The sigmoid, the unguarded calibrated law and the excited potential alone all reach it. With the guard and matched integrated excitation, no collapse occurs and the termination rule is passed up to $15\,\mathrm{keV}$, the end of our partial sweep (1–$15\,\mathrm{keV}$), whereas the sigmoid's series ends at $10\,\mathrm{keV}$. The guard also removes the reduction of surviving damage. Controls show why: $\mathcal{U}_{ht}$ alone reproduces the reduction, and injecting the excitation heat into ground-state cascades does not. The reduction comes from the excited potential acting outside its training domain in the cascade core.

1 Introduction

A $14.1\,\mathrm{MeV}$ fusion neutron can give a tungsten atom tens of keV. The resulting displacement cascade melts a few hundred cubic nanometres for a few picoseconds, and it drives the local electronic subsystem far from equilibrium. Classical molecular dynamics (MD) with a Born–Oppenheimer potential leaves the excited electrons out. Two-temperature models put them back as a heat bath but keep the ground-state forces [1–3]. Electron-temperature-dependent potentials change the forces, but they need a T_e-resolved fit and an electronic heat equation to supply T_e [4–6]. Wood, Koknat and Thompson recently proposed a cheaper middle way, a *layered* potential [7]. Two machine-learned potentials are fitted at a cold and a hot electronic smearing temperature. Every atom i then moves under the force blend $(1-\lambda_i)\mathcal{U}_{gs} + \lambda_i \mathcal{U}_{ht}$, with λ_i computed on the fly from the local ionic temperature T_i. In tungsten cascades this gave fewer surviving Frenkel pairs and smaller thermal spikes, but it made the dynamics unstable when the switching was aggressive.

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Everything physical in this construction passes through the switching law $\lambda(T_i)$. The original work used a logistic step at an ionic temperature $T^* \geq 0.5\,\mathrm{eV}$ and stated that its form was not optimised. This paper derives a switching law from three stated requirements and tests what it can achieve. The first is that the blend reproduce DFT across electronic temperature. We use a Sommerfeld-limited map $\Lambda(\theta) = 1 - e^{-(\theta/\theta_c)^2}$, whose single scale is fixed by a minimax-regret fit to DFT at the held-out temperature $T_e = 0.1\,\mathrm{eV}$. The second is that the electronic temperature seen by a cascade atom follow from a quasi-static two-temperature balance, $\theta = \kappa\,\Delta T_i$, with the ionic excess censored at the measured equilibrium noise of the T_i estimator. Together these give *calibrated Sommerfeld switching* (CSS), a smooth law with one physical coupling κ. The third requirement, added after the instability analysis, is that an atom feel the excited layer only where that layer lies inside its training domain. A neighbourhood trust guard enforces it, using the per-atom energy difference of the two layers as the domain certificate.

Our central finding is that these requirements can be met together, but that the damage reduction does not survive them. In detail:

1. **DFT calibration.** CSS lowers the force error at $0.1\,\mathrm{eV}$ by 13% relative to every published sigmoid, and lowers both energy and force errors integrated over T_e. Its energy error at $0.1\,\mathrm{eV}$ is 3% *higher*. The DFT data show why no single blend can satisfy both channels. The $0.1\,\mathrm{eV}$ surface lies only 0.045 (energies) to 0.082 (forces) of the way to the $0.5\,\mathrm{eV}$ surface, close to the Sommerfeld ratio 0.04. The much larger "optimal" blends reflect cancellation between the two fits' errors.

2. **An energy ledger.** The excitation work W_{ex} (Eq. 8) shows that force blending *heats* cascades by 6–8% of the recoil energy. It does so through a definite hysteresis: the spike disorders on the excited surface and re-orders after de-excitation. It is therefore not a form of electronic stopping.

3. **The instability mechanism.** Runaways are collapses into a spurious dense state of $\mathcal{U}_{ht}$, which lies $29\,\mathrm{eV/atom}$ below its own crystal at a density far outside its training data. The sigmoid, CSS and $\mathcal{U}_{ht}$ alone all reach it. A per-atom trust guard removes the collapse but makes the blend strongly non-conservative. The neighbourhood guard removes it while lowering the excitation work. At matched integrated excitation, it extends the stable recoil range under the original termination rule from $5\,\mathrm{keV}$ (sigmoid and unguarded CSS) to at least $15\,\mathrm{keV}$, the end of our sweep.

4. **The origin of the damage reduction.** At matched integrated excitation, the guarded law no longer reduces the surviving damage, and injecting the excitation heat into ground-state cascades does not reproduce the reduction either. $\mathcal{U}_{ht}$ alone does reproduce it. The reduction is produced by the excited layer in the out-of-domain core of the spike.

The switching law, its selection criteria and the sweeps were fixed in a time-stamped plan before any simulation. Every later change was declared, with its reason, before the runs that depend on it, and every intermediate variant is reported.

2 Related work

Electrons in radiation-damage MD. Electronic effects enter classical cascade simulations in two established ways. *Electronic stopping* applies a velocity-proportional drag to fast ions above a cutoff energy; the friction picture goes back to Caro and Victoria [8]. The *two-temperature model* (2T-MD) adds an electronic heat equation on a grid that exchanges energy with the ions through electron–phonon coupling [1, 2]. In tungsten, 2T-MD reduces the molten volume and the surviving damage of high-energy cascades [3], and stopping-based models show similar trends [9, 10]. Langevin baths with spatially correlated noise derive the coupling more carefully [11]. All of these keep the Born–Oppenheimer surface fixed and treat electrons as a heat bath. Electron-temperature-dependent potentials change the surface itself, through a T_e-dependent embedded-atom model for W [4] coupled to 2T-MD [5], through EAM extensions for laser-irradiated metals [12], or through neural-network potentials that take T_e as an input [6]. They require a T_e-resolved training set and, in practice, a 2T-MD solver to supply T_e. The electronic heat capacity and coupling of W are known to deviate strongly from free-electron forms at elevated T_e [13].

Layered (two-state) potentials. Wood, Koknat and Thompson [7] avoid a T_e grid. They fit two ACE potentials [14, 15] at two smearing temperatures and blend their forces atom by atom, with weights computed on the fly from the local ionic temperature. They found fewer surviving Frenkel pairs and smaller thermal spikes than with the ground-state potential alone. They also found unstable dynamics, with growing spike volumes after adaptive time stepping, when the switching temperature T^* was low, and they attributed this to long residency on the excited surface combined with the non-conservative force blend. Their switching law was chosen by physical intuition, and they state that its form and parameters were not optimised. At the intermediate temperature $0.1\,\mathrm{eV}$, their supplementary data already show that the best blend for energies differs from the best blend for forces.

Extrapolation of machine-learned potentials. Machine-learned potentials are reliable only near their training data. Active-learning schemes therefore monitor extrapolation on the fly, through the D-optimality grade of linear and ACE models [16, 17] or the disagreement of a model committee [18]. For radiation damage the critical region is the short-range, high-density part of the potential. Potentials for W are trained against dimer, short-distance and liquid data and joined to a screened repulsion for that reason [19]. Layered potentials double the exposure, because the excited layer must also be valid wherever the switching law puts weight on it. To our knowledge this has not been checked.

Switching between potentials. Interpolating between two force fields with a position- or state-dependent weight is common in adaptive-resolution MD and in QM/MM. When forces rather than energies are interpolated, energy is not conserved and an external thermostat has to absorb a systematic drift. This is analysed for AdResS [20] and repaired by a Hamiltonian formulation with a compensating drift term (H-AdResS [21]). In QM/MM the same distinction appears as force mixing versus energy mixing [22, 23]. Adaptive-precision ACE/EAM potentials use smooth polynomial switching functions of a structural descriptor [24], and a recent Hamiltonian version makes them conservative [25]. The present work keeps the non-conservative force blend fixed. Instead it asks which electronic-temperature map a DFT reference supports, and it measures the work done by the blend directly through Eq. (8). To our knowledge, neither a DFT calibration of the blend fraction across electronic temperatures nor an energy ledger of excitation work has been used for layered potentials.

3 Calibrated Sommerfeld switching

3.1 Layered dynamics and what a switching law must decide

Two interatomic potentials are fitted to density-functional theory (DFT) at a small and a large electronic smearing temperature: $\mathcal{U}_{gs}$ at $T_e = 0.001\,\mathrm{eV}$ and $\mathcal{U}_{ht}$ at $T_h = 0.5\,\mathrm{eV}$. The layered dynamics of Ref. [7] gives every atom a weight $\lambda_i \in [0, 1]$ and blends the two *forces*,

$$\mathbf{F}_i = -(1-\lambda_i)\nabla_i\mathcal{U}_{gs} - \lambda_i\nabla_i\mathcal{U}_{ht} = -\nabla_i\mathcal{U}_{gs} - \lambda_i\nabla_i\Delta\mathcal{U}, \qquad \Delta\mathcal{U} \equiv \mathcal{U}_{ht} - \mathcal{U}_{gs}. \tag{1}$$

We keep this algebra, both potentials and the per-atom LAMMPS implementation fixed. The only object we design is the map from the local state of an atom to λ_i. That map has to settle two separate questions. *(i) Which blend represents electrons at temperature θ?* This is a property of the potential-energy surfaces and can be checked against DFT. *(ii) What electronic temperature θ_i does an atom in a cascade see?* This is a property of the electron–ion energy exchange. The sigmoid used in Ref. [7], $\lambda_i = \left[1 + e^{-(T_i - T^*)/\sigma}\right]^{-1}$ with $\sigma = T^*/14$, answers both at once with one step at an ionic temperature $T^* \geq 0.5\,\mathrm{eV}$. Read as a map of temperature, it assigns $\lambda \approx 0$ to electrons at $0.1\,\mathrm{eV}$ and $\lambda \leq 0.5$ to electrons at the $0.5\,\mathrm{eV}$ training temperature of $\mathcal{U}_{ht}$ itself. Calibrated Sommerfeld switching (CSS) answers the two questions separately: $\lambda_i = \Lambda(\theta_i)$.

3.2 The electronic-temperature map $\Lambda(\theta)$

In the finite-temperature (Mermin) formulation of DFT [26], ions move on the electronic free energy $F(\mathbf{R}; T_e) = E - T_e S_e$. The Sommerfeld expansion [27] gives $F(\mathbf{R}; T_e) - F(\mathbf{R}; 0) = -\frac{\pi^2}{6} k_B^2 T_e^2\, g(\varepsilon_F; \mathbf{R}) + \mathcal{O}(T_e^4)$. Suppose the electronic correction separates into a configuration factor and a temperature factor, $F - F_0 = -s(\mathbf{R})H(T_e)$. Then the linear blend $\mathcal{U}_{gs} + \Lambda\,(\mathcal{U}_{ht} - \mathcal{U}_{gs})$ reproduces $F(\mathbf{R}; T_e)$ exactly when $\Lambda = H(T_e)/H(T_h)$. The Sommerfeld limit fixes $\Lambda \propto T_e^2$ as

$T_e \to 0$, the same physics Ackland built into a pairwise electronic-entropy term [28]. We adopt the one-parameter form

$$\Lambda(\theta) = 1 - \exp\left[-(\theta/\theta_c)^2\right], \tag{2}$$

which (a) reduces to the Sommerfeld law $(\theta/\theta_c)^2$ at low θ, (b) saturates at one so the blend never extrapolates beyond $\mathcal{U}_{ht}$, and (c) has a single scale θ_c. That scale is fixed by DFT at the intermediate temperature $T_m = 0.1\,\mathrm{eV}$, which neither potential was trained on: $\theta_c = T_m/\sqrt{-\ln(1-\lambda^*)}$.

Selection criterion for λ^*. At T_m we evaluate the uniform blend for $\lambda \in \{0, 0.025, \ldots, 1\}$ against all converged DFT configurations, in two error channels. The force channel is the RMSE over Cartesian force components. The energy channel is the per-atom energy RMSE after removing the mean per-atom residual of each set ("offset-free"). The offset is removed because each DFT set carries its own reference shift (FitSNAP's `ESHIFT`, Section 4) and the isolated-atom energy of both potentials is zero, so a constant per-atom offset carries no information about the surface. The energy-optimal and force-optimal fractions differ (Section 5), so no single blend minimises both. We therefore take the minimax relative regret, i.e. the point of the energy–force Pareto front where both channels exceed their own optimum by the same factor,

$$\lambda^* = \arg\min_{\lambda} \max \left\{ \frac{\mathrm{RMSE}_E(\lambda)}{\min_{\lambda'} \mathrm{RMSE}_E}, \frac{\mathrm{RMSE}_F(\lambda)}{\min_{\lambda'} \mathrm{RMSE}_F} \right\}. \tag{3}$$

The criterion is unit-free and has no tunable weight. It was fixed, together with Eq. (2), in a time-stamped plan before any potential was evaluated.

3.3 The driver: electronic excess temperature from the local ionic temperature

The cascade driver is the local atomic temperature T_i of Ref. [7]: the kinetic temperature of the atoms within $5.28\,\text{Å}$ of atom i, after removing their centre-of-mass velocity (LAMMPS `compute ave/sphere/atom`). To turn it into an electronic temperature we take the quasi-static limit of a local two-temperature balance [1, 2, 13]. Write $C_e\dot{T}_e = G(T_i - T_e) - K(T_e - T_b)$, with electron–phonon coupling G, a lumped electronic heat-removal rate K, and the bath temperature T_b. Its stationary solution is linear in the ionic excess,

$$\theta_i \equiv T_e - T_b = \kappa\,(T_i - T_b), \qquad \kappa = \frac{G}{G+K} \in (0, 1]. \tag{4}$$

The single dimensionless constant κ sets how aggressively the cascade samples $\mathcal{U}_{ht}$. $\kappa = 1$ is the fully coupled (local-equilibrium) upper bound. The ionic excess is measured above the statistical noise of the estimator. At the bath temperature, T_i fluctuates with standard deviation σ_T; we measured $\sigma_T = 130\,\mathrm{K}$ at $1000\,\mathrm{K}$ over $101 \times 54{,}000$ samples. We replace $T_i - T_b$ by the softly censored excess

$$\Delta T_i = s\,\mathrm{softplus}\left(\frac{T_i - T_f}{s}\right), \qquad T_f = T_b + 6\sigma_T, \quad s = \sigma_T. \tag{5}$$

Combining Eqs. (2)–(5),

$$\boxed{\lambda_i = 1 - \exp\left[-\left(\frac{\Delta T_i}{T_c}\right)^2\right], \qquad T_c = \frac{\theta_c}{\kappa}} \tag{6}$$

The law is C^∞ and monotone in T_i. It leaves the onset with zero slope ($\lambda \simeq (\Delta T_i/T_c)^2$), so noise in the estimator near the floor barely moves λ. It tends to 1 in the hot limit and to 0 in cold equilibrium. Its shape has no free parameter: the Gaussian form comes from the Sommerfeld limit, the scale θ_c from DFT, and the floor from the measured statistics of T_i. The 10–90% rise spans a factor of 4.7 in ΔT_i, against 1.37 in T_i for the sigmoid.

3.4 Matching aggressiveness

Any switching law trades how much of the cascade samples $\mathcal{U}_{ht}$ against how hard it does so. To compare CSS with a sigmoid of given T^* on equal terms, we pick κ so that both laws give the same integrated excitation of out-of-equilibrium atoms,

$$\chi = \int \mathrm{d}t \sum_{i:\,T_i > 2000\,\mathrm{K}} \lambda_i(t). \tag{7}$$

We use a frozen-driver estimate. Every run records histograms of $\log_{10} T_i$ (250 bins) at every fifth step during adaptive stepping and every 25 fs afterwards. From those, χ can be evaluated for *any* law on the same frames. T_c is then found by bisection so that CSS reproduces the sigmoid's χ on the sigmoid's own reference runs at 5 keV. The restriction to $T_i > 2000\,\mathrm{K}$ ($8\sigma_T$ above the bath) removes a bath-level tail that would otherwise dominate χ. For $T^* = 0.5\,\mathrm{eV}$ the sigmoid assigns $\lambda(1000\,\mathrm{K}) = 9.3 \times 10^{-6}$ to every atom, which adds ≈ 1 atom of permanent excitation per 10^5 atoms, independent of the cascade.

3.5 Excitation work: an energy ledger for force blending

Equation (1) gives an exact energy identity. With K the ionic kinetic energy and E_{th} the cumulative energy exchanged with thermostats,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(K + \mathcal{U}_{gs} + E_{\mathrm{th}}\big) = -\sum_i \lambda_i\, \mathbf{v}_i \cdot \nabla_i \Delta\mathcal{U} \equiv P_{\mathrm{ex}}, \qquad W_{\mathrm{ex}}(t) = \int_0^t P_{\mathrm{ex}}\,\mathrm{d}t'. \tag{8}$$

W_{ex} is the energy that the excited-state forces put into ($W_{\mathrm{ex}} > 0$) or take out of ($W_{\mathrm{ex}} < 0$) the ground-state system. Physically, the electronic subsystem should absorb energy from a cascade [8, 1, 11]. For a uniform and constant λ, $W_{\mathrm{ex}} = -\lambda[\Delta\mathcal{U}(t) - \Delta\mathcal{U}(0)]$ is a state function and the dynamics are conservative. For a single excursion of a region with $\lambda : 0 \to 1 \to 0$, $W_{\mathrm{ex}} = \Delta\mathcal{U}(\mathrm{on}) - \Delta\mathcal{U}(\mathrm{off})$. Excitation therefore *heats* the ions whenever the region relaxes on $\mathcal{U}_{ht}$ toward configurations where $\mathcal{U}_{ht}$ is relatively low and only then switches back. This is the same bookkeeping that governs force-interpolating adaptive-resolution and QM/MM schemes [20–23]. We measure W_{ex} at every step from `compute pair` energies of the unscaled sub-styles.

3.6 A domain guard for the excited layer

A layered potential is only as trustworthy as its excited layer, and $\mathcal{U}_{ht}$ is a fit to DFT data of near-equilibrium density. Section 7 shows that cascades drive atoms far outside that domain, into a spurious collapsed state of $\mathcal{U}_{ht}$, and that this is where the dynamics run away. The final law therefore adds a guard, $\lambda_i = \Lambda(\theta_i)\, G_i$, which keeps an atom on $\mathcal{U}_{gs}$ whenever its force depends on an excited-layer site energy outside the training domain. The domain certificate is the per-atom energy difference $\Delta e_i = e_i^{ht} - e_i^{gs}$ of the two ACE expansions, which the blend evaluates anyway. With $g(\Delta e) = [1 + e^{-(\Delta e - \Delta_{\mathrm{lo}})/w}]^{-1}$,

$$G_i = g(\Delta e_i) \prod_{j \neq i} \Big[1 - \big(1 - g(\Delta e_j)\big)\, s(r_{ij})\Big], \tag{9}$$

where s tapers from 1 at the cutoff of $\mathcal{U}_{ht}$ (4.68 Å) to 0 at 5.18 Å. The product runs over neighbours because ACE is many-body: the force on atom i contains $-\partial e_j/\partial \mathbf{r}_i$ for every site j within the cutoff. Both parameters come from the training data of the two potentials. $\Delta_{\mathrm{lo}} = -0.447\,\mathrm{eV}$ is the 0.1st percentile of Δe over both training sets (median $0.58\,\mathrm{eV}$). $w = 0.16\,\mathrm{eV}$ is twice the robust spread of the same distributions. G_i is evaluated inside the force call, lagged by one step, at no measurable cost. We call the guarded law CSS-gn. The guard does not change the accuracy at the anchors: at $0.1\,\mathrm{eV}$ the blended errors are 13.01 meV/atom and 0.0804 eV/Å, against 13.00 and 0.0803 without it. That is because only 1.3% of the atoms in the DFT configurations have $g < 0.99$. The guard was introduced after the instability analysis of Section 7, together with a per-atom variant that failed (Section 7). Its definition and pass criteria were fixed before any cascade was run with it.

Matching a guarded law. The guard removes excitation that the frozen-driver estimate of Eq. (7) cannot see. We therefore re-matched κ for CSS-gn empirically. On guarded runs at a trial coupling, $r = \chi_{\mathrm{hot}}^{\mathrm{realized}}/\chi_{\mathrm{hot}}^{\mathrm{frozen}}$ measures the fraction the guard removes. The coupling is then chosen so that $r\,\chi_{\mathrm{hot}}^{\mathrm{frozen}}(T_c)$ equals the sigmoid's realized χ_{hot} at 5 keV. Two iterations, fixed in advance as final, gave $T_c = 4464\,\mathrm{K}$ ($\kappa = 0.50$), against $\kappa = 0.30$ for the unguarded CSS-m(0.5).

4 Computational protocol

Potentials and DFT reference. $\mathcal{U}_{gs}$ and $\mathcal{U}_{ht}$ are the published linear ACE potentials of Ref. [7], each overlaid with ZBL repulsion between 3.927 and 4.488 Å, used unmodified. The DFT reference

comprises the corresponding VASP sets (PBE, PAW W_sv_GW, 700 eV) at $T_e = 0.001$, 0.1 and 0.5 eV. After excluding every configuration marked as not converged they contain 9467, 2444 and 9445 configurations. The JSON field `Energy` is the internal energy E and is the field FitSNAP [29] reads as the energy target. We found that `FreeEng` is VASP's $E(\sigma \to 0) = (E + F)/2$, so the Mermin free energy is $F = 2\,\texttt{FreeEng} - E$. At 0.5 eV only F satisfies $\mathrm{d}F/\mathrm{d}V = -P$ against the DFT stresses of the equation-of-state set (residual 9×10^{-4} eV/Å^3, against 5×10^{-2} eV/Å^3 for E). The primary energy label is therefore E, as in the fits, and F is a secondary label. Model energies and forces were obtained with LAMMPS [30] (`pace product`, exactly as distributed) after rotating each cell into LAMMPS' triclinic convention with FitSNAP's QR procedure. The MD uses `pace recursive`, which evaluates the same basis and agrees to within 6×10^{-11} eV (energies) and 3×10^{-12} eV/Å (forces) on all 2444 configurations of the 0.1 eV set.

Implementation of the switching laws. Every law is an atom-style LAMMPS variable of `c_lavg[2]` (`compute ave/sphere/atom`, cutoff 5.28 Å). It feeds `pair_style hybrid/scaled 1.0 zbl ... v_wgs pace ... v_lam pace`, so all laws use the same driver and the same per-atom force blend, Eq. (1). A small patch makes `hybrid/scaled` evaluate $\mathcal{U}_{ht}$ only on atoms whose weight is at least $\varepsilon = 10^{-4}$, plus their neighbours. Forces on those atoms are bit-identical to the unpatched code; elsewhere they deviate by at most $\varepsilon|\mathbf{F}|$ (measured 2×10^{-4} eV/Å). The patch makes a blended step 2.4 times faster, about the cost of $\mathcal{U}_{gs}$ alone. Energies of the sub-styles, which `hybrid/scaled` does not tally for atom-style weights, are read from `compute pair`. Re-evaluating the variables from the t_0 snapshots of 7 runs reproduces the weights used in the dynamics to within 9×10^{-6}.

Cascades. We follow the cascade protocol of Ref. [7] and its published LAMMPS input. BCC W is equilibrated at 1000 K and 1 atm (lattice parameter 3.1873 Å for $\mathcal{U}_{gs}$, 3.1653 Å for EAM) and thermalised by NVT in cubic boxes. The primary knock-on atom (PKA) is the atom nearest the box centre and receives the recoil energy in an isotropic random direction. Repetition r at energy E uses the same random seed for every law, so all comparisons are paired over identical PKA directions. Atoms within 2.5% of the box length from a face are thermostatted at 1000 K (Nosé–Hoover, 0.1 ps [31]); the interior is NVE. Adaptive time stepping (`fix dt/reset`, 10^{-6}–0.5 fs, maximum displacement 0.05 Å) continues until the step returns to 0.5 fs; this instant is t_0. After t_0 the whole cell is held at 1000 K. A blended run switches to $\mathcal{U}_{gs}$ alone once the hottest atom has cooled below 2000 K. The Marinica EAM [32] is run with the same protocol. Within our 32-core budget we used smaller boxes than Ref. [7]: 20^3, 25^3, 30^3 and 40^3 BCC cells (1.6×10^4 to 1.28×10^5 atoms) at 1, 2, 5 and 10–15 keV, i.e. 8–16 atoms per eV of recoil energy against 25–250 in the original work. Frenkel pairs are counted 4 ps after t_0 instead of 30 ps. Both deviations apply identically to every law; the end of this section isolates the effect of the anneal length. The protocol of Ref. [7] calls for 25 PKA directions per law and energy. We ran 25 for $\mathcal{U}_{gs}$, the $T^* = 0.5$ eV sigmoid, CSS-m(0.5), CSS-gn and EAM at 1–10 keV, and for $\mathcal{U}_{gs}$, CSS-gn and EAM at 15 keV. The two unguarded laws have six directions at 15 keV, above the energy at which their series had already terminated. Secondary laws and controls have fewer directions at some energies; every point with fewer than 25 is marked as partial in Tables 2 and 3. The sweep stops at 15 keV: by our measured cost model a 20 keV point would need a 50^3 box at about 30 core-hours per cascade. It therefore covers 1–15 keV of the original 1–200 keV.

Controls. Besides $\mathcal{U}_{gs}$-only and EAM cascades we ran two controls. In $\mathcal{U}_{ht}$-only cascades, $\mathcal{U}_{ht}$ is used everywhere, on its own crystal equilibrated with the same protocol ($a = 3.1914$ Å at 1000 K). In heated ground-state cascades, a $\mathcal{U}_{gs}$-only run receives at t_0, as kinetic energy of the atoms with $T_i > 2000$ K (uniform velocity scaling), exactly the excitation work that the paired sigmoid run accumulated by $t_0 + 4$ ps. Up to t_0 these runs are bit-identical to the $\mathcal{U}_{gs}$ runs.

Stability rule. The original study terminates a recoil series once more than 25% of the repetitions show a thermal-spike volume that increases after adaptive time stepping. We apply this rule literally, with one operational detail fixed on control runs only. In our runs the non-crystalline volume of every cascade, including ground-state and EAM cascades, keeps growing for a few hundred femtoseconds after t_0. A rule that also compares $V(t_0 + 0.5\,\mathrm{ps})$ with $V(t_0)$ therefore flags 46 of 250 control runs ($\mathcal{U}_{gs}$ only and EAM), where no instability exists. A repetition is therefore flagged when $\max\{V(t_0{+}1), V(t_0{+}2), V(t_0{+}4\,\mathrm{ps})\} > V(t_0)$, or when adaptive stepping fails to terminate

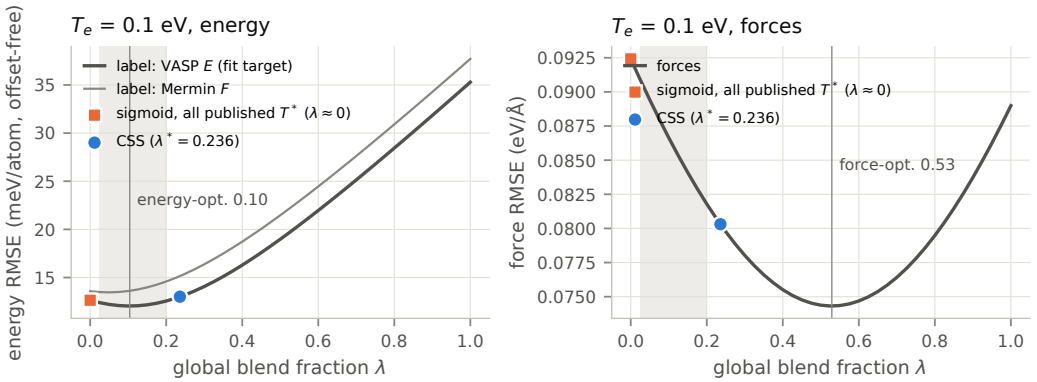


Figure 1: Uniform-blend error against the held-out DFT set at $T_e = 0.1\,\mathrm{eV}$ (2444 configurations). Left: offset-free energy RMSE for the fitted label E (dark) and the Mermin free energy F (light). Right: force RMSE. Squares mark the sigmoid of Ref. [7] ($\lambda \approx 0$ at $0.1\,\mathrm{eV}$ for every published T^*); circles mark CSS at $\lambda^* = 0.236$. The shaded interval contains every blend that improves on $\lambda = 0$ in both channels.

within $10\,\mathrm{ps}$. This rule flags 1 of the 250 control runs. We also record *collapse events*, runs in which an atom re-accelerates above $300\,\mathrm{eV}$ after t_0, when the time step is already fixed at $0.5\,\mathrm{fs}$. No control run exceeds $106\,\mathrm{eV}$, no run of a guarded law exceeds $169\,\mathrm{eV}$, and every collapse event exceeds $704\,\mathrm{eV}$ (Fig. 9d). Collapse events are reported alongside the termination rule but do not enter it.

Analysis. The thermal-spike volume is computed with OVITO [33]. Atoms not identified as BCC by polyhedral template matching [34] are enclosed by an alpha-shape surface mesh [35] with probe radius $1\,\mathrm{nm}$, and the enclosed volume is taken. As in Ref. [7], the peak spike volume is the volume at t_0; we also evaluate it at $t_0 + 0.5$, 1, 2 and $4\,\mathrm{ps}$. Frenkel pairs are counted by Wigner–Seitz analysis against the pre-PKA configuration. χ_{hot} (Eq. 7), the deep residency $\chi_{>0.5} = \int N(\lambda_i > 0.5)\,\mathrm{d}t$, and W_{ex} (Eq. 8) are integrated from per-step records during adaptive stepping and records every $5\,\mathrm{fs}$ afterwards. Means are given with standard errors, intervals are bootstrap 95% confidence intervals over repetitions, paired differences use the Wilcoxon signed-rank test, and flagged fractions carry Clopper–Pearson intervals. A collapse event produces up to several hundred spurious Frenkel pairs. Damage statistics are therefore reported both over all repetitions and over repetitions without a collapse event, together with the number of such events. Paired differences are taken over directions without a collapse event in either run.

Anneal length. To isolate the shorter anneal, we continued 4 cascades at $1\,\mathrm{keV}$ of $\mathcal{U}_{gs}$, the sigmoid, CSS-m(0.5) and CSS-gn on the same directions to $t_0 + 30\,\mathrm{ps}$. Between 4 and $30\,\mathrm{ps}$ the mean Frenkel-pair count falls from 3.00 to 2.00 for $\mathcal{U}_{gs}$, stays at 1.75 and 1.25 for the two unguarded laws, and falls from 3.75 to 3.25 for CSS-gn. The paired differences to $\mathcal{U}_{gs}$ shrink accordingly, from -1.25 to -0.25 (sigmoid), from -1.75 to -0.75 (CSS-m) and from $+0.75$ to $+1.25$ (CSS-gn) pairs per cascade. The $4\,\mathrm{ps}$ read-out therefore overstates the damage reduction of the unguarded laws by up to one Frenkel pair per cascade at this energy, without changing the order of the laws.

5 Accuracy against DFT across electronic temperature

Energy- and force-optimal blends disagree strongly. Figure 1 reproduces the observation of Supplemental Fig. 18 of Ref. [7] under a reference-free energy metric, and sharpens it. At $0.1\,\mathrm{eV}$ the energy-optimal uniform blend is $\lambda_E = 0.104$, the force-optimal blend is $\lambda_F = 0.529$, and the minimax-regret criterion selects $\lambda^* = 0.236$. This gives $\theta_c = 0.193\,\mathrm{eV}$ (2237 K) in Eq. (2). The two anchors then follow without further input: $\Lambda(0.001\,\mathrm{eV}) = 2.7 \times 10^{-5}$ and $\Lambda(0.5\,\mathrm{eV}) = 0.9988$. The energy optimum depends on the reference convention: anchoring energies to the lowest BCC structure gives $\lambda_E = 0.097$, and a single cohesive-energy shift gives $\lambda_E = 0.067$. The force optimum does not.

Table 1: Blended-potential error against DFT at the three electronic temperatures, with λ taken from each switching law read as a map of temperature. Energies: offset-free per-atom RMSE (meV/atom), label E. Forces: component RMSE (eV/Å). "Integrated" is the trapezoidal average over $T_e \in [0.001, 0.5]$ eV through the three anchors. CSS assigns $\lambda = 2.69 \times 10^{-5}$, 0.236 and 0.999 at the three anchors. Grouped 5-fold cross-validation at 0.1 eV holds out whole configuration families; there λ^* is refitted on the training folds and the energy offset is taken from the training folds.

	T_e = 0.001 eV		T_e = 0.1 eV (held out)		T_e = 0.5 eV		integrated	
law	E	F	E	F	E	F	E	F
CSS	42.23	0.6368	13.00	**0.0803**	**31.32**	**0.2273**	**23.24**	**0.1944**
sigmoid T^* = 0.5 eV	42.23	0.6368	**12.64**	0.0924	32.23	0.3221	23.43	0.2385
sigmoid T^* = 0.75 eV	42.23	0.6368	12.64	0.0924	46.47	0.4790	29.13	0.3014
sigmoid $T^* \geq$ 1.25 eV	42.23	0.6368	12.64	0.0924	46.80	0.4821	29.27	0.3026
grouped CV at 0.1 eV: CSS			14.75	0.0804	(λ^* refits: 0.141–0.327)			
grouped CV at 0.1 eV: sigmoid			12.83	0.0924				

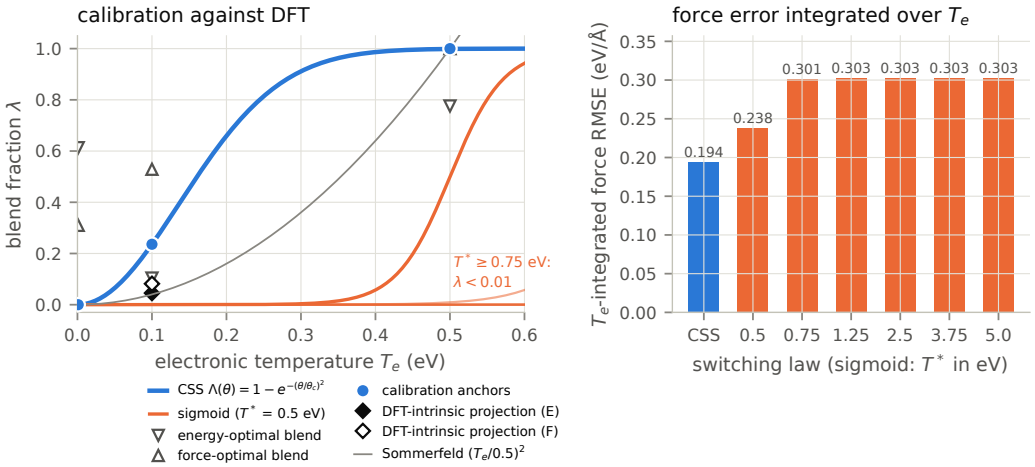


Figure 2: Left: calibration of λ against electronic temperature. Triangles are the energy- and force-optimal uniform blends at each DFT temperature. Circles are the CSS anchors $(0, 0)$, $(0.1, \lambda^*)$, $(0.5, 1)$. Diamonds are the DFT-intrinsic projection of the 0.1 eV distortion onto the 0.5 eV distortion (geometry-matched configurations), and the thin grey curve is the Sommerfeld law $(T_e/T_h)^2$. Right: force RMSE integrated over $T_e \in [0.001, 0.5]$ eV.

CSS against the sigmoid. Table 1 compares the two laws at the anchors. Every published value $T^* \in [0.5, 5]$ eV maps 0.1 eV electrons to $\lambda < 10^{-4}$, that is, to the ground-state surface. CSS lowers the force RMSE at 0.1 eV from 0.0924 to 0.0803 eV/Å. The improvement survives grouped cross-validation: CSS has lower force error in 3 of the 4 held-out families that contain non-zero forces. At the 0.5 eV anchor the sigmoid assigns $\lambda = 0.5$ ($T^* = 0.5$ eV) or $\lambda \approx 0$ (every larger T^*) to electrons at the training temperature of $\mathcal{U}_{ht}$, which nearly doubles the force error there. Integrated over T_e, CSS lowers both channels relative to every published T^*. CSS does *not* lower the energy error at the held-out point: 13.00 against 12.64 meV/atom in-sample, and 14.75 against 12.83 in cross-validation. The minimax criterion trades a 3% energy loss for a 13% force gain. All blends that would have improved both channels lie in $0 < \lambda \lesssim 0.2$ (shaded in Fig. 1). We report this as a failure of the pre-registered selection against the strict criterion and analyse its cause next.

What the optimal blend measures: error cancellation, not electronic physics. The optimal uniform blend at an electronic temperature can be computed at the training temperatures as well as at the intermediate one (Fig. 2). On the DFT data that $\mathcal{U}_{gs}$ was fitted to ($T_e = 0.001$ eV), the error is lowest not at $\lambda = 0$ but at $\lambda_F = 0.310$ for forces and $\lambda_E = 0.613$ for energies. On the 0.5 eV data of $\mathcal{U}_{ht}$, the energy optimum is $\lambda_E = 0.776$. Physically the correct fractions at the training temperatures are 0 and 1. The offsets therefore measure partial cancellation between the independent fitting errors

Table 2: Main results. DFT errors at the held-out $T_e = 0.1\,\mathrm{eV}$ (offset-free energy RMSE and force RMSE); for $\mathcal{U}_{ht}$ only this is the uniform blend $\lambda = 1$. At 5 keV: realized excitation χ_{hot}, excitation work W_{ex} at $t_0 + 4\,\mathrm{ps}$, and the paired change of surviving Frenkel pairs relative to $\mathcal{U}_{gs}$ only (runs without collapse). Flagged runs under the termination rule of Ref. [7], with collapse events in parentheses; bold marks more than 25%, which ends the series. †Partial point: fewer than the 25 directions of the protocol. Our sweep covers 1–15 keV of the original 1–200 keV, so "≥ 15" is right-censored at our highest energy. n.a.: not defined for this law (χ_{hot} when $\lambda \equiv 1$; W_{ex} without the $\mathcal{U}_{gs}$ ledger; DFT error of EAM, which is not a blend of the two layers). [a]Read from the damage figure of Ref. [7] (30 ps anneal, larger boxes). [b]Highest recoil energy of the $T^* = 0.5\,\mathrm{eV}$ series in Ref. [7].

	DFT, $T_e = 0.1\,\mathrm{eV}$		5 keV cascades			flagged/runs (collapses)			stable
law	E meV/at.	F eV/Å	χ_{hot} atom·ps	W_{ex} eV	ΔFP %	5 keV	10 keV	15 keV	up to keV
$\mathcal{U}_{gs}$ only	12.64	0.0924	0	0	0	0/25	1/25	0/25	≥ 15
sigmoid $T^* = 0.5\,\mathrm{eV}$, as reported [7]	*not reported*				-55[a]	*counts not reported*			15[b]
sigmoid $T^* = 0.5\,\mathrm{eV}$, reproduced	12.64	0.0924	589	403	-45	3/25 (1)	**15/25 (8)**	**3/6 (3)**†	5
CSS-m(0.5), matched κ	13.00	0.0803	550	321	-53	4/25 (1)	**11/25 (6)**	1/6 (1)†	5
CSS-κ1, $\kappa = 1$	13.00	0.0803	2488	353	-55	4/25 (1)	*not run*		≥ 5
CSS-gn (guarded, matched)	13.01	0.0804	594	183	-11	1/25	3/25	0/25	≥ 15
CSS-g (per-atom guard)	13.01	0.0804	667	1375	-46	0/11†	*not run*		≥ 5
SIG-gn (guarded sigmoid)	12.64	0.0924	214	189	-10	0/8†	1/8†	0/6†	≥ 15
$\mathcal{U}_{ht}$ only	35.29	0.0890	n.a.	n.a.	-49	2/12 (2)†	1/8 (1)†	**3/6 (3)**†	10
EAM (Marinica)	n.a.	n.a.	0	n.a.	$+26$	0/25	0/25	0/25	≥ 15

of the two potentials. $\mathcal{U}_{gs}$, fitted with a large energy weight, has a force RMSE of 0.6368 eV/Å on its own training data, against 0.4292 for the best blend. To isolate the physical distortion we used the DFT data alone. 1891 configurations occur with identical geometry in all three sets. For these we projected the $0.1\,\mathrm{eV}$-minus-$0.001\,\mathrm{eV}$ change onto the $0.5\,\mathrm{eV}$-minus-$0.001\,\mathrm{eV}$ change. The projection coefficient is 0.045 for relative energies (0.049 with the Mermin label) and 0.082 for forces. The rms force change at $0.1\,\mathrm{eV}$ is only $0.0158\,\mathrm{eV/Å}$, against $0.115\,\mathrm{eV/Å}$ at $0.5\,\mathrm{eV}$. These numbers are close to the Sommerfeld ratio $(0.1/0.5)^2 = 0.04$. The true surface distorts monotonically and almost quadratically in T_e, as Eq. (2) assumes at small θ. The large and mutually inconsistent "optimal" blends ($\lambda_E \approx 0.1$, $\lambda_F \approx 0.5$) come mostly from model error. A single global blend cannot fix model error, so no blend can be simultaneously energy- and force-optimal; this is a property of the two fits, not of the electrons. We also tested an alternative explanation, that the mismatch arises because the fits use the internal energy E although the forces are derivatives of the Mermin F. With the thermodynamically consistent label the energy optimum moves *further* from the force optimum ($\lambda_E = 0.048$), so this explanation is refuted.

6 Cascade dynamics

Table 2 collects the main results; the rest of this section and the next two explain them.

Behaviour at equilibrium. In a $1000\,\mathrm{K}$ crystal without a recoil (16,000 atoms, $18\,\mathrm{ps}$), CSS-κ1, the most aggressive CSS setting, never raised any atom above $\lambda = 3.2 \times 10^{-3}$. Only a fraction 7×10^{-7} of atom-samples exceeded 10^{-3}, and the total excitation was $\eta = \sum_i \lambda_i = 0.0024$. The $T^* = 0.5\,\mathrm{eV}$ sigmoid kept every atom below $\lambda = 7 \times 10^{-5}$. Its tail value at the bath temperature, $\lambda(1000\,\mathrm{K}) \approx 9 \times 10^{-6}$, nevertheless sums to a permanent $\eta = 0.147$. Neither law toggles atoms under thermal fluctuations. In CSS this follows from the censoring floor and the zero slope at onset (Fig. 3). The guard of Eq. (9) is inactive in the crystal. Even in cascade snapshots, 99.9% of the atoms with $T_i < 2000\,\mathrm{K}$ have $\Delta e_i > 0.38\,\mathrm{eV}$, far above Δ_{lo}.

Matched excitation, different residency. CSS-m(0.5) uses $T_c = 7349\,\mathrm{K}$ ($\kappa = 0.30$). The frozen-driver estimator reproduces the sigmoid's realized χ_{hot} to 0.15% on the reference runs. Over all $5\,\mathrm{keV}$ repetitions without a collapse event, the realized excitation is $\chi_{\mathrm{hot}} = 589 \pm 17$ (sigmoid, $n = 25$) against 550 ± 11 atom·ps (CSS-m(0.5), $n = 25$). The re-matched guarded law reaches 594 ± 12 atom·ps ($n = 25$). At matched excitation, CSS spends far less time on the excited surface itself. The deep residency $\chi_{>0.5} = \int N(\lambda_i > 0.5)\,\mathrm{d}t$ is 513 ± 16 for the sigmoid against

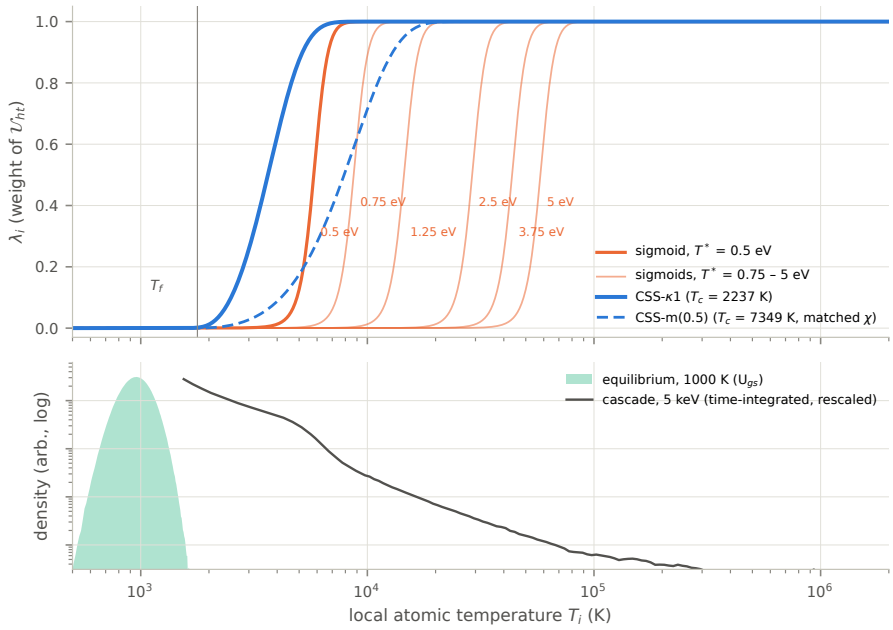


Figure 3: Top: switching laws as functions of the local atomic temperature. The sigmoids of Ref. [7] at the six published values of T^* are shown in orange, CSS in blue. CSS-κ1 is the parameter-free fully coupled law ($T_c = \theta_c$). CSS-m(0.5) is matched to the $T^* = 0.5\,\mathrm{eV}$ sigmoid in integrated excitation χ_{hot} at 5 keV. Bottom: the equilibrium distribution of T_i at 1000 K ($101 \times 54{,}000$ samples, shaded) and the time-integrated T_i distribution of 5 keV cascades (line). The censoring floor $T_f = T_b + 6\sigma_T$ lies above every equilibrium sample.

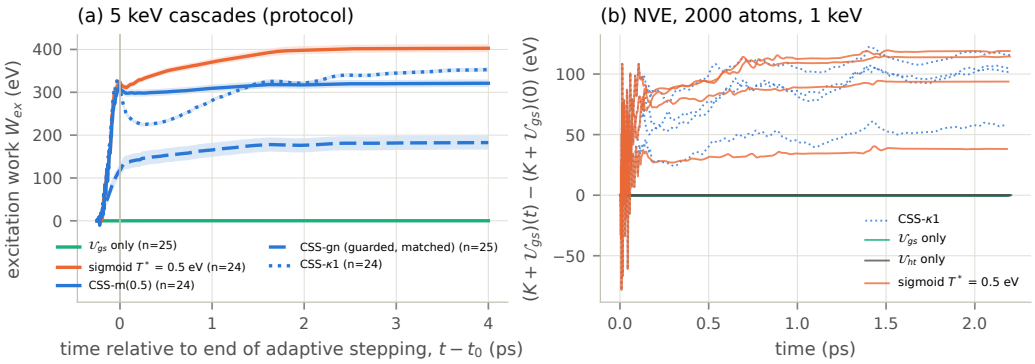


Figure 4: Excitation work W_{ex}, Eq. (8). (a) 5 keV cascades under the full protocol, mean ± s.e.m. over repetitions without a collapse event, against time relative to the end of adaptive stepping. (b) Fully microcanonical 1 keV cascades in 2000 atoms, four PKA directions each. Ground-state-only and $\mathcal{U}_{ht}$-only dynamics conserve their own energies to within 1 eV. Both blended laws add energy.

225 ± 5 atom·ps for CSS, a factor of 2.3. CSS spreads the same excitation as partial weights over the warm, still solid periphery of the spike.

The force blend heats cascades. For ground-state-only dynamics W_{ex} stays within 1 eV of zero (integration error), and for $\mathcal{U}_{ht}$-only dynamics $K + \mathcal{U}_{ht}$ is conserved to the same level. Every blended run instead *gains* energy (Fig. 4). In the microcanonical 2000-atom test at 1 keV, $W_{\mathrm{ex}} = +91 \pm 19$ eV for the sigmoid and $+95 \pm 13$ eV for CSS-κ1. At 2 keV in 16,000 atoms the values are $+160 \pm 11$ and $+144 \pm 10$ eV. Under the full protocol at 5 keV they are 403 ± 10 eV (sigmoid) and 321 ± 9 eV (CSS-m(0.5)) by $t_0 + 4$ ps, that is 6–8% of the recoil energy. The sign is opposite to that of electronic stopping or electron–phonon coupling, which remove energy from the ions [8, 1, 3]. The layered force blend therefore does not act as a drag.

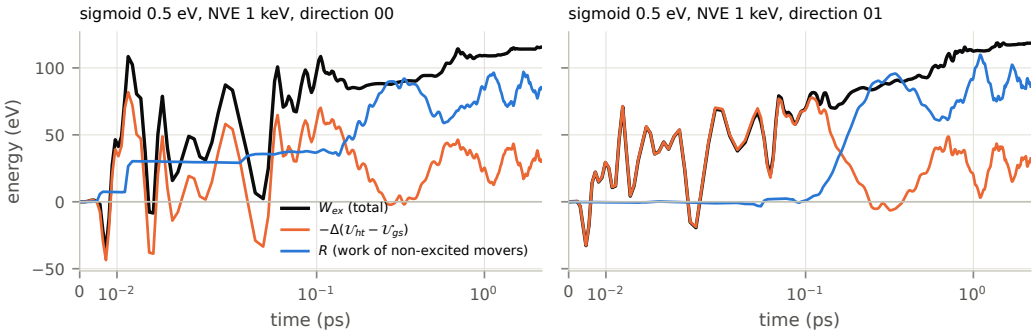


Figure 5: Decomposition of the excitation work in two microcanonical 1 keV cascades with the full $\mathcal{U}_{ht}$ evaluated on every atom. $W_{\mathrm{ex}} = -\Delta(\mathcal{U}_{ht} - \mathcal{U}_{gs}) + R$, where $R = \int \sum_i (1-\lambda_i)\mathbf{v}_i \cdot \nabla_i \Delta\mathcal{U}\,\mathrm{d}t$ is the work of moving atoms that are not fully excited.

Where the energy comes from. Equation (8) splits exactly into $W_{\mathrm{ex}} = -[\Delta\mathcal{U}(t) - \Delta\mathcal{U}(0)] + R(t)$. R vanishes as long as every moving atom is fully excited, which is the case in the ballistic phase. A recoiling atom raises the centre-of-mass-corrected temperature of every sphere it passes through, so it and all its neighbours sit at $\lambda \approx 1$. The ballistic work is therefore a state function: it equals the drop of $\mathcal{U}_{ht} - \mathcal{U}_{gs}$ as the excited region is disordered, to within 2 eV at 0.1 ps (Fig. 5). The disordered, compressed configurations of the spike core are where the two potentials disagree most. There $\mathcal{U}_{ht}$ lies lower relative to $\mathcal{U}_{gs}$ than it does in the crystal, by 62–70 eV over a 1 keV spike. In the thermal phase the region re-orders and $\Delta\mathcal{U}$ recovers. By then most atoms have been released to the ground-state surface, so this recovery is never charged to the ions, and R grows by the same amount. The net heating is a hysteresis: the spike disorders while excited and re-orders after de-excitation. The ballistic part is set by the hottest atoms, where every law saturates at $\lambda = 1$, so it hardly depends on the law: 324 ± 8 against 310 ± 9 eV at t_0 at 5 keV. The thermal-phase part does depend on the law. A law that keeps partial excitation while the spike re-orders pays part of $\Delta\mathcal{U}$ back. Between t_0 and $t_0 + 4$ ps, CSS-m(0.5) adds 11 ± 4 eV, against 79 ± 7 eV for the sigmoid.

Damage and spike volume. Both unguarded laws reproduce the damage reduction reported for layered dynamics (Fig. 6). In paired comparisons over the same PKA directions, the sigmoid changes the Frenkel-pair count by -35%, -42% and -45% at 1, 2 and 5 keV ($p = 0.0039$, 0.0011, 1×10^{-4}, Wilcoxon). CSS-m(0.5) changes it by -38%, -41% and -53% ($p = 0.0044$, 0.0011, 4×10^{-5}).

At 10 keV, where both series terminate, neither change is significant: over the 17 and 19 directions without a collapse event it is $+13\%$ for the sigmoid ($p = 0.36$) and -16% for CSS-m(0.5) ($p = 0.26$). The 8 and 6 collapse events among their 25 directions each leave hundreds of spurious Frenkel pairs, so all-run means are no damage measure there. Beyond the end of both series, the six directions run at 15 keV (a partial point) give -61% and -39%. The peak spike volume $V(t_0)$ is smaller with both unguarded laws at 5 keV (-18% and -20%; $p = 0.0022$ and 1×10^{-4}) and at 10 keV (-18% and -17%; $p = 0.0093$ and 0.0062). Part of this difference reflects when adaptive stepping ends rather than the size of the spike. Blended cascades reach t_0 earlier, and their non-BCC region decays more slowly afterwards (Fig. 7): between $t_0 + 0.5$ ps and $t_0 + 2$ ps it is larger than in $\mathcal{U}_{gs}$ cascades. The guarded law CSS-gn, at the same integrated excitation, does not reduce the damage at any energy. Its changes are -5%, -14%, -11% and -5% at 1–10 keV (none significant). At 15 keV the change is -13% ($p = 0.45$, $n = 25$). Section 8 explains why.

The sigmoid family against the original study. Figure 8 compares the reproduced sigmoid at three of the published T^* with the values reported for them. At 1–5 keV every reported damage reduction lies within our 95% interval at the same T^* and energy, with one exception. At 5 keV we find -45%, -41% and -27% for $T^* = 0.5$, 1.25 and 5 eV, against -55%, -41% and -12% reported. The exception is $T^* = 5$ eV at 5 keV (our interval $[-39, -16]\%$), where the reported value also stands out from the reductions of 21–26% reported at 1–4 keV. The excitation scales alike with T^*: relative to the $T^* = 0.5$ eV series at 5 keV, χ is 0.135 and 0.019 for $T^* = 1.25$ and 5 eV, against 0.103 and 0.019 in the original. The changes of the peak spike volume agree at 5 keV to within about one standard error (-7.6, -8.0 and -7.6 nm^3, against -10, -10 and -8 nm^3 reported), and are

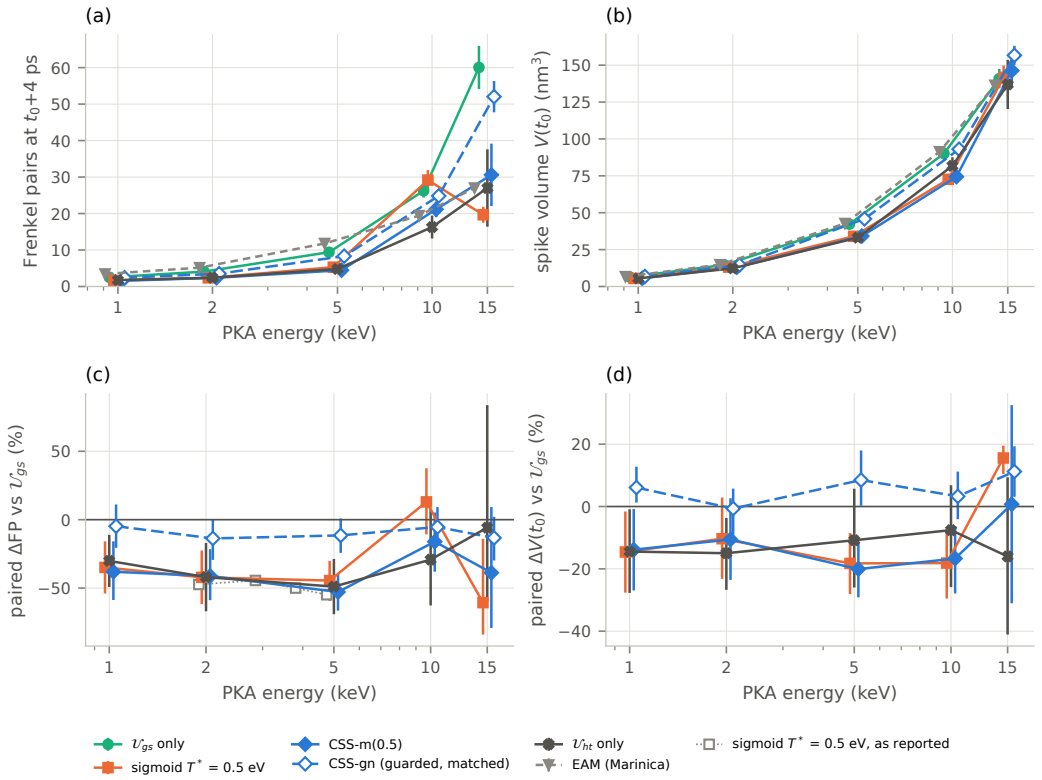


Figure 6: Surviving damage and spike volume across the recoil spectrum. (a) Frenkel pairs 4 ps after t_0 and (b) non-BCC volume at t_0: means ± s.e.m. over repetitions without a collapse event. (c, d) Paired changes relative to $\mathcal{U}_{gs}$-only cascades on the same PKA directions, with bootstrap 95% intervals. Grey squares in (c) are the relative changes of the $T^* = 0.5$ eV sigmoid read from the damage figure of Ref. [7] (30 ps anneal, larger boxes).

smaller than reported at 10 keV (-16.1, -15.1 and -15.6 nm^3, against -25, -34 and -26 nm^3). The clearest gap is at 10 keV. There no reproduced sigmoid reduces the damage ($+13\%$, $+7\%$ and -4%, none significant), whereas the original reports -47% and -13% for $T^* = 1.25$ and 5 eV. Our ground-state cascades also retain more damage at that energy (26.2 Frenkel pairs, against about 15 in the original), consistent with our shorter anneal (4 ps instead of 30 ps) in a smaller box.

Stability along the recoil series. Table 3 and Fig. 9 apply the termination rule of Ref. [7] to every law. Our sweep is partial: it covers 1–15 keV of the original 1–200 keV, so a stable range that reaches 15 keV is right-censored there. EAM is never flagged, and $\mathcal{U}_{gs}$ only once in 25 directions at 10 keV, by a spike at most 2% larger after t_0 and without a collapse event. The reproduced $T^* = 0.5$ eV sigmoid passes 1–5 keV but fails at 10 keV: 15/25 directions are flagged, 8 of them by a collapse event. Its stable range in our protocol is therefore 5 keV, against 15 keV in the original study with its larger boxes and longer anneal. The pre-registered CSS-m(0.5) fails at the same energy (11/25, 6 collapse events), so spreading the same excitation over a broader temperature range does not stabilize the dynamics. CSS-gn, at the same realized excitation at 5 keV, stays below the threshold at every energy (3/25 at 10 keV, 0/25 at 15 keV) and has no collapse event in any run. That extends the stable range at matched excitation from 5 keV to at least 15 keV, the end of our sweep. The Clopper–Pearson 95% upper bounds on its flagged fraction are 31% and 14% at 10 and 15 keV; the lower bounds for the sigmoid and CSS-m(0.5) at 10 keV are 39% and 24%. The guarded sigmoid behaves the same way (1/8 and 0/6, partial points), so the guard, not the CSS shape, is what stabilizes. At larger T^* the reproduced sigmoid needs no guard within our sweep (Fig. 9b, c). The $T^* = 1.25$ and 5 eV series pass every energy up to 15 keV (1/6 and 1/6 flagged there; partial points), consistent with the 50 and ≥ 200 keV reported for them. The two studies thus order the stable ranges alike in T^*; the reproduction is shorter at $T^* = 0.5$ eV (5 against 15 keV), and the longer original ranges lie

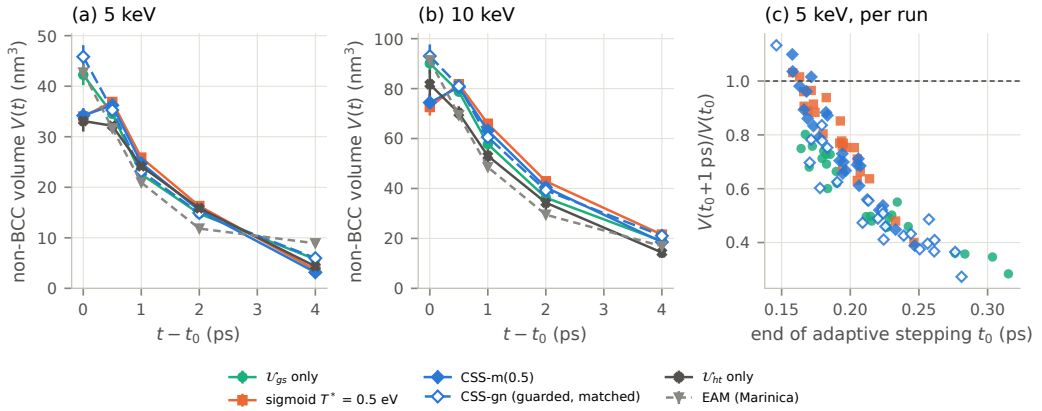


Figure 7: Time-resolved non-BCC volume after the end of adaptive stepping, means over repetitions without a collapse event, at (a) 5 keV and (b) 10 keV. (c) Ratio $V(t_0{+}1\,\mathrm{ps})/V(t_0)$ against t_0 for every 5 keV run. A run above the dashed line is flagged by the termination rule.

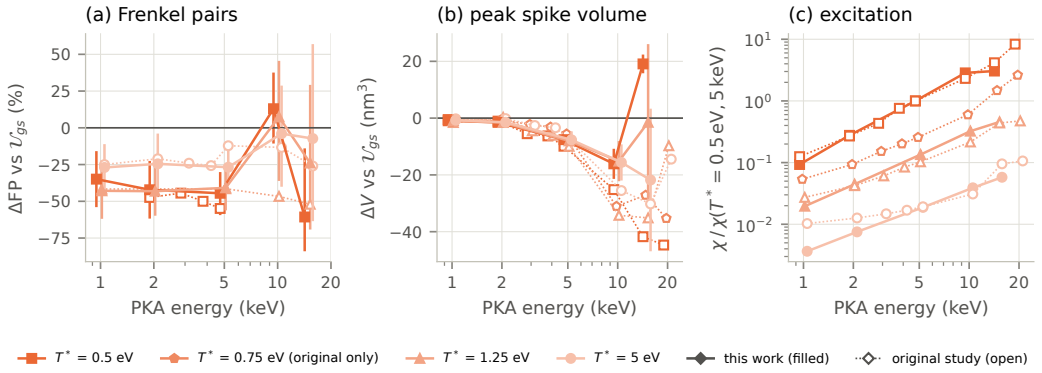


Figure 8: The reproduced sigmoid family against the original study [7], per recoil energy (filled: this work, runs without a collapse event; open: values reported in the original). (a) Paired change of surviving Frenkel pairs relative to $\mathcal{U}_{gs}$, with bootstrap 95% intervals; original from its Fig. 7. (b) Change of the peak spike volume relative to $\mathcal{U}_{gs}$, paired mean $\pm$ s.e.m.; original from its Fig. 6, whose points are plotted against χ and are assigned to recoil energies through the $\chi(E)$ series of its Fig. 5. (c) Integrated excitation relative to the $T^* = 0.5\,\mathrm{eV}$ series at 5 keV in each study; original from its Fig. 5 (the two studies report χ in different units). Original values were read from the published figures by colour segmentation of the markers. Our $T^* = 1.25$ and 5 eV points at 10 and 15 keV are partial (8 and 6 directions).

beyond our sweep. These sigmoids are stable because they excite little: at 5 keV their χ_{hot} is 79 and 11 atom·ps, against 589 for $T^* = 0.5\,\mathrm{eV}$ and 594 for CSS-gn. Even so, the $T^* = 1.25\,\mathrm{eV}$ sigmoid has 1 collapse event in its 6 directions at 15 keV. CSS-gn is the only law in this study that stays stable at the excitation of the $T^* = 0.5\,\mathrm{eV}$ sigmoid. The flagged runs fall into two groups (Fig. 9d). Collapse events, the $\mathcal{U}_{ht}$ artefact of Section 7, account for most flags at 10–15 keV: 29 runs in total, all of them with an atom above 704 eV. The rest, including the flagged runs of CSS-gn and SIG-gn and the single $\mathcal{U}_{gs}$ run, show mild late growth of the non-BCC region. In blended runs that region decays more slowly after t_0 (Fig. 7), so a spike that ends adaptive stepping early can still be larger 1 ps later.

7 Why blended cascades run away, and a trust guard

A spurious collapsed state of $\mathcal{U}_{ht}$. Under isotropic compression of BCC W, $\mathcal{U}_{gs}$ rises steeply, by $20\,\mathrm{eV/atom}$ at a lattice parameter of 2.60 Å (Fig. 10a). $\mathcal{U}_{ht}$ rises by only $4.6\,\mathrm{eV/atom}$ up to the same compression. It then falls into a minimum of $-37\,\mathrm{eV/atom}$ at $a = 2.15$ Å (nearest-neighbour

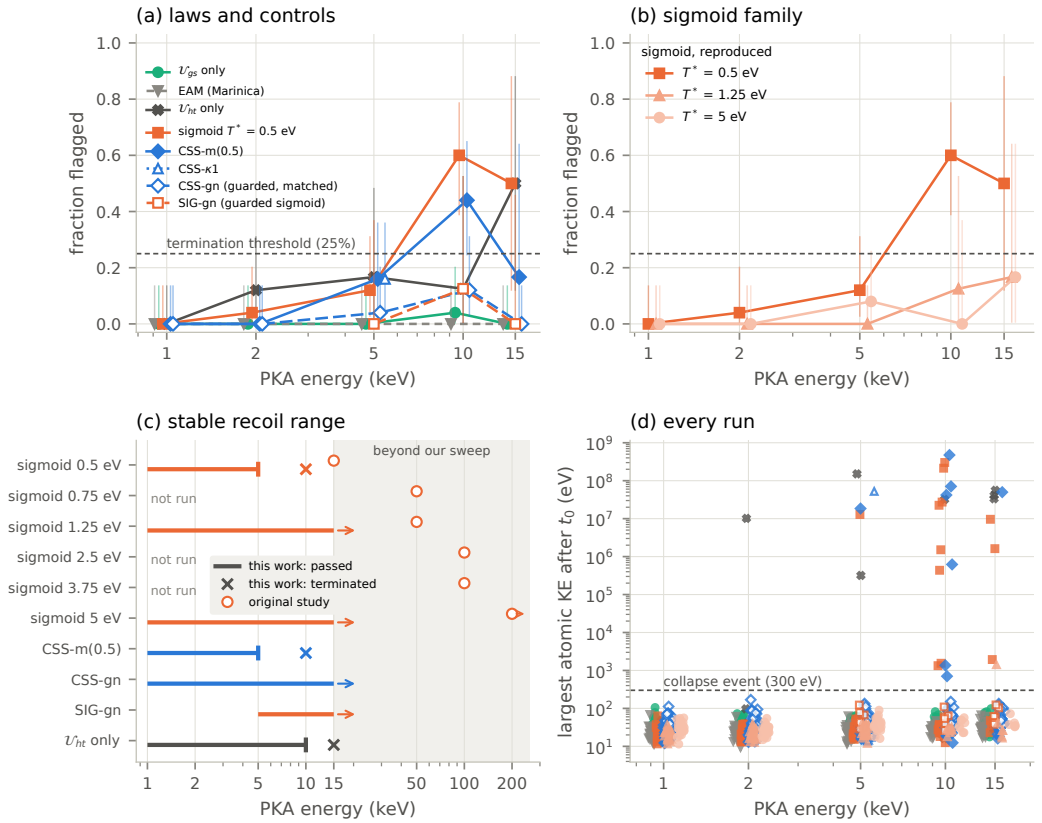


Figure 9: Stability along the recoil series (partial sweep, 1–15 keV). (a, b) Fraction of repetitions flagged by the termination rule (spike volume larger at t_0+1, 2 or 4 ps than at t_0), with Clopper–Pearson 95% intervals, for (a) the laws of this work and the controls and (b) the reproduced sigmoid at $T^* = 0.5$, 1.25 and 5 eV. A series terminates at the first energy above 25% (dashed). (c) Stable recoil range: energies passed in this work (bars; an arrow marks a series not terminated within our sweep, a cross the energy at which it terminates) and the highest recoil energy of each T^* series in Fig. 5 of Ref. [7] (open circles; for $T^* = 0.5$ eV from its text). (d) Largest atomic kinetic energy reached after t_0 in every run. Values above 300 eV are collapse events: re-acceleration at a fixed 0.5 fs step, which no control run approaches.

distance 1.86 Å), 29 eV/atom *below* its own equilibrium crystal. The collapsed state has an atomic volume of 5.0 Å^3, against 16.1 Å^3 at equilibrium. The densest configuration in the 0.5 eV training set has 14.7 Å^3 per atom and no interatomic distance below 1.92 Å. The minimum is therefore an extrapolation artefact of the many-body expansion of $\mathcal{U}_{ht}$, not electronic physics. The isolated W dimer shows no such minimum, so the artefact is many-body.

How a cascade reaches it. We re-ran the unstable CSS-m(0.5) trajectory at 5 keV with the same domain decomposition. The re-run reproduced the original diagnostics bit for bit, and this time we recorded dense snapshots. 1.20 ps after t_0, 16 atoms in a warm, locally compressed region already have $e_i^{ht} - e_i^{gs}$ below the training-domain floor, the 0.1st percentile of this difference over the training data of both potentials (−0.447 eV; median 0.58 eV). The most extreme of them has $e_i^{ht} - e_i^{gs} = -29.4$ eV ($e_i^{ht} = -32.4$ eV, $e_i^{gs} = -3.0$ eV) at $\lambda_i = 0.34$ and $T_i = 6513$ K (Fig. 10b). With that weight, the blended force pulls the atoms further into the collapsed state. The energy released heats the neighbourhood, which raises the weights of the neighbours and pulls more atoms in. By $t_0 + 1.35$ ps, 55 atoms are outside the training range and the fastest atom carries 272 eV. A keV-scale secondary cascade follows, disorders most of the box, and leaves 438 Frenkel pairs. Throughout, the time step is already fixed at 0.5 fs, and the excitation work of the run grows to the order of 10^7 eV.

Table 3: Flagged repetitions per recoil energy under the termination rule; collapse events in parentheses. Bold: more than 25% flagged, which terminates the series. †Partial point: fewer than the 25 directions of the protocol; the rule is applied to the directions run. The sweep covers 1–15 keV of the 1–200 keV of Ref. [7], so "≥ 15" is right-censored at our highest energy. Last column: highest recoil energy of each T^* series in Fig. 5 of Ref. [7]; [a]quoted from its text (the figure shows the series to 20 keV); [b]the end of its sweep. n.a.: no counterpart in the termination study of Ref. [7]. CSS-gn and SIG-gn carry the neighbourhood guard of Eq. (9); CSS-gn is matched to the $T^* = 0.5$ eV sigmoid in realized $\chi_{\rm hot}$ at 5 keV.

	flagged/runs (collapses)					stable up to (keV)	
law	1 keV	2 keV	5 keV	10 keV	15 keV	this work	Ref. [7]
sigmoid of Ref. [7], $\sigma = T^*/14$, reproduced							
$T^* = 0.5$ eV	0/25	1/25	3/25 (1)	**15/25 (8)**	**3/6 (3)**†	5	15[a]
$T^* = 0.75$ eV				*not run*			50
$T^* = 1.25$ eV	0/25	0/25	0/25	1/8†	1/6 (1)†	≥ 15	50
$T^* = 2.5$ eV				*not run*			100
$T^* = 3.75$ eV				*not run*			100
$T^* = 5$ eV	0/25	0/25	2/25	0/8†	1/6†	≥ 15	≥ 200[b]
this work and controls							
CSS-m(0.5)	0/25	0/25	4/25 (1)	**11/25 (6)**	1/6 (1)†	5	n.a.
CSS-κ1	0/1†	0/1†	4/25 (1)	*not run*		≥ 5	n.a.
CSS-g	*not run*		0/11†	*not run*		≥ 5	n.a.
CSS-gn	0/25	0/25	1/25	3/25	0/25	≥ 15	n.a.
SIG-gn	*not run*		0/8†	1/8†	0/6†	≥ 15	n.a.
$\mathcal{U}_{ht}$ only	0/25	3/25 (1)	2/12 (2)†	1/8 (1)†	**3/6 (3)**†	10	n.a.
$\mathcal{U}_{gs}$ only	0/25	0/25	0/25	1/25	0/25	≥ 15	n.a.
EAM (Marinica)	0/25	0/25	0/25	0/25	0/25	≥ 15	n.a.

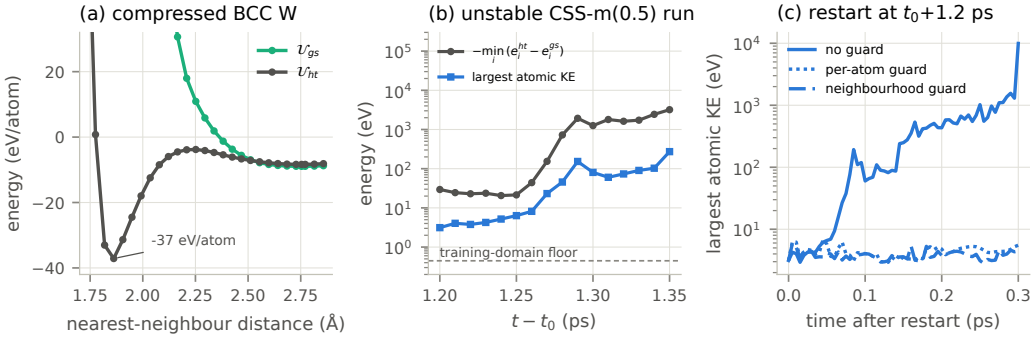


Figure 10: The runaway mechanism. (a) Energy per atom of BCC W under isotropic compression. $\mathcal{U}_{ht}$ has a spurious collapsed state far outside its training data. (b) Dense re-run of an unstable CSS-m(0.5) cascade (5 keV). The most negative per-atom energy difference $e_i^{ht} - e_i^{gs}$ already lies two orders of magnitude below the training-domain floor before the largest atomic kinetic energy starts to grow. (c) Restart from the state at $t_0 + 1.2$ ps of that trajectory, without a guard, with the per-atom guard and with the neighbourhood guard of Eq. (9).

The sigmoid fails the same way. The unstable sigmoid trajectory, re-run with dense snapshots, reproduces its original diagnostics exactly as well. There the collapse starts in the hot core immediately after adaptive stepping. At t_0, 163 atoms already lie below the training floor; their median weight is $\lambda_i = 1.00$ and their median local temperature 9500 K. The most negative energy difference drops from -7 eV at t_0 to -196 eV at $t_0 + 0.15$ ps, while the largest atomic kinetic energy starts to grow. Both laws therefore fail through the same artefact of $\mathcal{U}_{ht}$. They reach it by different routes: the sigmoid puts the whole hot core on $\mathcal{U}_{ht}$, whereas CSS puts partial weight on long-lived warm regions.

The excited potential collapses on its own. The force blend is not needed for the collapse. Cascades run entirely on $\mathcal{U}_{ht}$, which conserve $K + \mathcal{U}_{ht}$, collapse from 2 keV upwards: 1 of 25 directions at 2 keV, 2 of 12 at 5 keV, 1 of 8 at 10 keV and 3 of 6 at 15 keV. At 15 keV $\mathcal{U}_{ht}$ alone

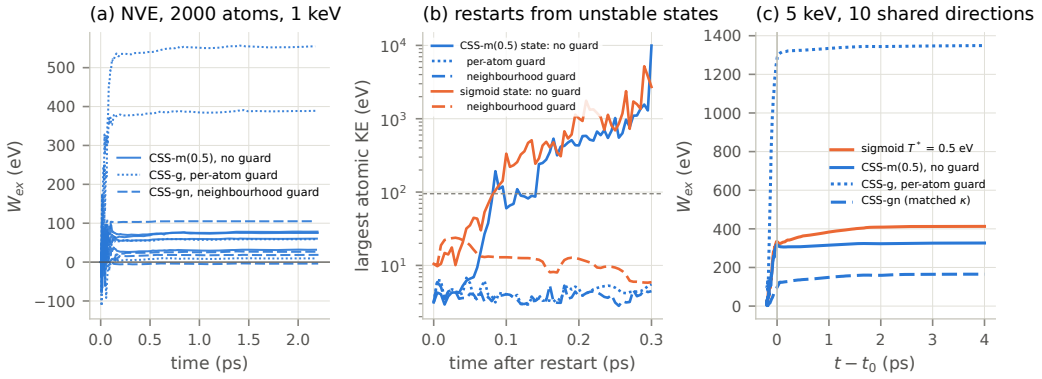


Figure 11: Per-atom against neighbourhood guard. (a) Excitation work in microcanonical 1 keV cascades (2000 atoms, $\mathcal{U}_{ht}$ evaluated everywhere, four PKA directions) for CSS-m(0.5) without a guard, with the per-atom guard (CSS-g) and with the neighbourhood guard. (b) Largest atomic kinetic energy after restarts from the two unstable states. (c) Mean excitation work of 5 keV cascades over the PKA directions shared by the four laws (runs without collapse).

therefore fails the termination rule (3/6 flagged, a partial point), and its stable range is 10 keV. In the first collapsing run at 5 keV the potential energy drops by 9.3×10^4 eV within 0.5 ps of t_0, as the dense phase nucleates in the hot core and grows. A phase that lies 29 eV/atom below the crystal and is reached across a barrier of only 4.6 eV/atom acts as a sink that nothing in the dynamics bounds. The force blend adds its own non-conservative energy to the event (Section 6), but it is not the root cause. The original attribution of the instability to long residency on $\mathcal{U}_{ht}$ and the non-conservative blend [7] is thus correct in its first part. That residency, however, is harmful because of where $\mathcal{U}_{ht}$ is evaluated.

Guarding the excited layer: per atom is not enough. Since the collapsed state lies far outside the training data, the natural remedy is to keep atoms off $\mathcal{U}_{ht}$ where $\mathcal{U}_{ht}$ is out of domain. Our first attempt applied the certificate of Section 3.6 per atom, $\lambda_i = \Lambda(\theta_i)\, g(\Delta e_i)$ ("CSS-g"). It removes the collapse. Restarted from the state at $t_0 + 1.2$ ps of the unstable trajectory, the largest atomic kinetic energy stays below 7 eV for 0.3 ps, whereas the unguarded law reaches 9942 eV (Fig. 10c). In 11 cascades at 5 keV it produced no flag and no runaway. However, it makes the force blend strongly non-conservative. In the microcanonical 1 keV test the excitation work is $+253 \pm 131$ eV, against $+61 \pm 11$ eV without the guard (Fig. 11a). At 5 keV it is 1309 ± 163 eV at t_0, against 310 ± 9 eV. The reason is the many-body structure of ACE. Switching off only the out-of-domain site j leaves its neighbours exposed to the spurious gradient $-\partial e_j^{ht}/\partial \mathbf{r}_k$, while j itself moves on $\mathcal{U}_{gs}$, so the pair forces no longer balance. The neighbourhood guard of Eq. (9) closes this gap. In the microcanonical test its excitation work is $+37 \pm 24$ eV, below that of the unguarded law. From the unstable CSS state it keeps the largest kinetic energy below 6 eV. From the unstable sigmoid state, the sigmoid with the same guard (SIG-gn) stays below 24 eV, whereas the bare sigmoid reaches 5201 eV. At t_0 of 5 keV cascades the neighbourhood guard switches off 39% of the atoms with $\Lambda > 0.01$ (at most 74%). The hot core of a cascade is thus largely outside the domain in which $\mathcal{U}_{ht}$ was fitted. Both guarded variants were designed after the instability had been observed and are reported as such. Their definitions and pass criteria were fixed before any cascade was run with them.

8 What reduces the damage

Both unguarded laws reduce the surviving damage at 5 keV. In paired comparisons over the same PKA directions, the Frenkel-pair count changes by -45% with the sigmoid ($p = 1 \times 10^{-4}$, Wilcoxon) and by -53% with CSS-m(0.5) ($p = 4 \times 10^{-5}$). With the guarded law at the same integrated excitation it changes by only -11% ($[-24, +1]\%$, $p = 0.11$, $n = 25$). Three explanations are possible, and two controls separate them (Fig. 12).

Not electronic dissipation. The original interpretation was inelastic scattering, i.e. energy loss to the electrons [7]. Section 6 shows that the force blend adds energy instead.

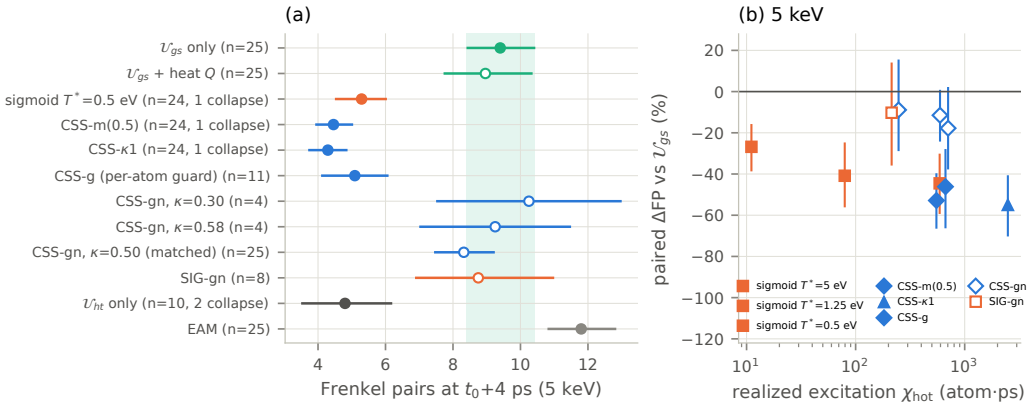


Figure 12: Frenkel pairs of 5 keV cascades for every law and control. (a) Mean and bootstrap 95% interval over repetitions without a collapse event; the band is the $\mathcal{U}_{gs}$-only interval. Open symbols: $\mathcal{U}_{gs}$ dynamics with the excitation work of the paired sigmoid run injected as heat at t_0 ("$\mathcal{U}_{gs}$ + heat Q"), and the guarded laws. (b) Paired change of the Frenkel-pair count relative to $\mathcal{U}_{gs}$ against the realized excitation χ_{hot}. Filled symbols are laws that leave the hot core on $\mathcal{U}_{ht}$: the unguarded laws, and CSS-g, whose per-atom guard still exposes the neighbours of out-of-domain sites. Open symbols are the neighbourhood-guarded laws. The reduction does not follow the amount of excitation. The $T^* = 5\,\mathrm{eV}$ sigmoid excites little more than the fastest recoils ($\chi_{\mathrm{hot}} = 11$ atom·ps) and changes the damage by -27%. CSS-gn, at $\chi_{\mathrm{hot}} = 594$ atom·ps, changes it by -11%.

Not the excitation heat. To test whether the added energy itself anneals the damage, we ran $\mathcal{U}_{gs}$-only cascades over the same 25 directions. At t_0 each received, as kinetic energy of its hot region, the excitation work that the paired sigmoid run had accumulated by $t_0 + 4\,\mathrm{ps}$. These runs are bit-identical to the $\mathcal{U}_{gs}$ runs up to t_0. The heat changes the Frenkel-pair count by only -5% ($[-14, +6]\%$, $p = 0.39$), much less than the sigmoid itself.

The excited surface, outside its domain. $\mathcal{U}_{ht}$ alone, with no blending at all, reproduces the reduction (-49%, $p = 0.0059$, $n = 12$ directions, 2 repetitions with a collapse event excluded). It does so across the lower half of our sweep: -30% and -42% at 1 and 2 keV ($p = 0.0089$ and 0.0066, over 25 and 24 directions without a collapse event), and -29% at 10 keV ($n = 7$, not significant; Fig. 6). The reduction therefore comes from the dynamics on the excited surface. The guard tells us where on that surface. It removes $\mathcal{U}_{ht}$ only from atoms whose force depends on an out-of-domain site, and at t_0 these are 39% of the excited atoms: the hot core. With the hot core on $\mathcal{U}_{gs}$ and the same total excitation placed on the in-domain periphery, the reduction disappears. The damage reduction of layered dynamics in W thus originates in the part of the spike where $\mathcal{U}_{ht}$ is evaluated outside its training domain. That is the compressed, disordered core, where $\mathcal{U}_{ht}$ is far softer than $\mathcal{U}_{gs}$ (Fig. 10a), the same regime in which it collapses. As fitted, the excited layer gives no DFT-backed reason to expect less damage. Whether a $\mathcal{U}_{ht}$ trained on compressed, electronically hot configurations would reproduce the effect is an open question that these potentials cannot answer.

9 Discussion

9.1 Where the pre-registered law fails, and why

Energy accuracy at the held-out temperature. CSS lowers the force error at 0.1 eV and both integrated errors, but not the energy error at 0.1 eV (13.00 against 12.64 meV/atom). The reason is structural, not a matter of tuning. The DFT reference changes by only 0.045–0.082 of its 0.5 eV change at 0.1 eV, close to Sommerfeld T_e^2 scaling. The two potentials, however, differ from DFT (and from each other) by much more than that physical signal. Their fitting errors are partly anti-correlated, so a global blend lowers the error even on the training data of either potential: $\lambda_F = 0.310$ at 0.001 eV and $\lambda_E = 0.776$ at 0.5 eV. Any calibration of a *blend fraction* against a *total* error therefore mixes physics with model error, and the energy- and force-optimal fractions need not agree. The DFT-intrinsic projection, a calibration against the DFT change itself, would give $\Lambda(0.1\,\mathrm{eV}) \approx 0.05$–$0.08$. That improves on the ground-state surface in both channels at 0.1 eV (Fig. 1, shaded band), but it

was not our pre-registered rule, so we report it only as a diagnostic. Two remedies follow, neither pursued here because both change the potentials. One is to fit $\mathcal{U}_{ht}$ as a correction to $\mathcal{U}_{gs}$ on the same configurations (Δ-learning), which removes the uncorrelated model errors from the blend. The other is to add DFT data at intermediate T_e, so that $\Lambda(\theta)$ is constrained by more than one interior point.

Stability of the pre-registered law. At matched integrated excitation, CSS-m(0.5) put far less weight on the fully excited surface than the sigmoid, which we expected to stabilize the dynamics. It did not. Both laws reach the same spurious collapsed state of $\mathcal{U}_{ht}$, by different routes: the sigmoid through the fully excited hot core, CSS through partially excited warm regions. Both pass $5\,\mathrm{keV}$ with one collapse in 25 runs each, and both are terminated at $10\,\mathrm{keV}$ (15/25 and 11/25 flagged). No reshaping of $\lambda(T_i)$ can avoid this. Any law that assigns weight to hot, compressed atoms exposes them to the artefact, and a law that does not do so no longer excites the cascade. The instability has to be addressed in configuration space, which is what the guard does.

The guard removes the damage reduction. The guarded law passes the termination rule up to $15\,\mathrm{keV}$ and, at matched excitation, adds less energy than either unguarded law. However, it no longer reduces the surviving damage. Section 8 traces the reduction to dynamics on $\mathcal{U}_{ht}$ in its out-of-domain region. The combination "stable, DFT-anchored and damage-reducing" is therefore not reachable by any switching law built on these two potentials. The stable law loses the effect because the effect, as these potentials produce it, is an extrapolation.

9.2 What this means for layered potentials

Layered force blending is not electronic stopping. Stopping and electron–phonon coupling remove energy from the ions, and 2T-MD lowers cascade damage in W through that dissipation [3, 9]. The layered blend instead adds 6–8% of the recoil energy, through excited disordering of the spike that is paid back only after de-excitation. The damage it removes is removed by the softness of the excited layer in a configuration range where it was never fitted. For a layered potential to make predictions about damage, the excited layer must be valid in the hot, compressed core of a cascade. That calls for T_e-resolved training data at high density, of the kind that machine-learned potentials for radiation damage already include for the ground state through short-range repulsion and dimer data [19]. It also calls for an on-the-fly measure of extrapolation, such as the per-atom energy difference used here or the extrapolation grade of ACE and related active-learning schemes [17, 16, 18]. A conservative alternative to force blending is to mix energies, with an explicit electronic reservoir that pays for every change of λ [25, 21]. That would make the excitation work a physical electronic energy rather than an artefact.

9.3 Limitations

(i) Protocol scale. With 32 cores we used smaller cubic boxes than the original study (8–16 instead of 25–250 atoms/eV) and counted Frenkel pairs $4\,\mathrm{ps}$ after adaptive stepping instead of $30\,\mathrm{ps}$. The isolating comparison in Section 4 bounds the effect of the anneal length on the paired differences. *(ii) Statistics and sweep.* The matched comparison has the protocol's 25 directions per energy, but several secondary points are partial (marked in Tables 2 and 3). Even at 25 directions the termination rule is a coarse test, so we give Clopper–Pearson intervals with every flagged fraction. The sweep ends at $15\,\mathrm{keV}$, the lower end of the original 1–$200\,\mathrm{keV}$; whether the guarded law stays stable at the higher energies, which the original sigmoids with $T^* \geq 0.75\,\mathrm{eV}$ reach, is untested. *(iii) The driver.* T_i is a 40–50-atom kinetic temperature with no memory. It cannot distinguish a fast atom crossing a sphere from a hot sphere, and it carries no electronic relaxation time. The quasi-static closure $\theta = \kappa\Delta T$ is the simplest consistent choice, not a derived electronic temperature. *(iv) One calibration point.* $\Lambda(\theta)$ is anchored at a single interior temperature, so its shape between anchors rests on the Sommerfeld argument rather than on data. *(v) Coupling.* A physical value of $\kappa = G/(G+K)$ for W would require the electronic heat-removal rate of a nanometre-sized spike. With $G \sim 10^{17}$–$10^{18}\,\mathrm{W\,m^{-3}K^{-1}}$ [13] and diffusive removal $K \approx \kappa_e \pi^2/R^2 \sim 10^{20}\,\mathrm{W\,m^{-3}K^{-1}}$ (electronic thermal conductivity $\kappa_e \sim 100\,\mathrm{W\,m^{-1}K^{-1}}$, spike radius $R \sim 3\,\mathrm{nm}$), the estimate is $\kappa \sim 10^{-3}$. That is two orders of magnitude below the matched values studied here, which suggests that all of these settings, including the original ones, overstate the electronic excitation of a W thermal spike. *(vi) The guard certificate.* Δe_i detects configurations in which the two layers disagree beyond their training spread. It cannot detect regions where both layers extrapolate in the same way. *(vii)*

Post-hoc components. The guard and the re-matching of the guarded law were introduced after the instability had been observed. Their definitions, parameters and pass criteria were fixed before the runs that test them, and every intermediate variant is reported.

10 Conclusion

We derived a switching law for layered excited-state potentials from three requirements. It should agree with DFT across electronic temperature, follow a quasi-static electron–ion balance, and never use the excited layer outside its training domain. The DFT calibration improves the force error at the held-out temperature and both integrated errors. It cannot improve energies and forces together there, because the two fits' errors exceed the physical signal. An exact energy ledger shows that force blending heats cascades rather than damping them. The instabilities that limit aggressive switching are collapses into a spurious dense state of the excited potential. A neighbourhood trust guard removes them and extends the stable recoil range at matched excitation from $5\,\mathrm{keV}$ to at least $15\,\mathrm{keV}$, the end of our sweep, with the protocol's 25 directions at every energy. The same guard removes the reduction of surviving damage. Together with $\mathcal{U}_{ht}$-only and heated ground-state controls, this places the origin of that reduction in the out-of-domain response of the excited potential in the cascade core, not in electronic physics supported by the training data. Layered potentials can become predictive for radiation damage once their excited layer is trained, and certified on the fly, in the dense, hot configurations of a cascade core.

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